9th Class • Punjab Board • PECTAA New Syllabus

Chapter 5: Work, Energy and Power

Complete solved exercise — work, kinetic and potential energy, conservation, power, efficiency and energy resources.

Chapter 5 • Complete official exercise • 9 MCQs + 10 short + 9 constructed + 5 comprehensive + 13 numericals

What this chapter is about

• **Work** W = Fd cos θ is energy transferred by a force through displacement. Its SI unit is the joule (J).
• **Kinetic energy** K = ½mv² depends on motion; **gravitational potential energy** U = mgh depends on height relative to a reference.
• **Power** P = W/t = E/t is the rate of doing work or transferring energy; 1 W = 1 J s^{-1}.
• **Efficiency** = useful energy or power output ÷ total input × 100%. Real machines lose energy to heat, sound and friction.

Conventions: Take g = 10 m s^{-2}. Work is scalar but can be positive, negative or zero. Torque also has unit N m, yet it is not the same physical quantity as work measured in joules.

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A. Multiple Choice Questions

1. Work done is maximum when the angle between the force F and the displacement d is:

(a) 0° (b) 30°
(c) 60° (d) 90°

Answer: (a) 0° — W = Fd cos θ. For a fixed force and displacement, cos θ is greatest at θ = 0°, so the full force acts along displacement.

2. A joule can also be written as:

(a) kg m s^{-2} (b) kg m s^{-1}
(c) kg m^2 s^{-3} (d) kg m^2 s^{-2}

Answer: (d) kg m^2 s^{-2} — 1 J = 1 N m = (kg m s^{-2})(m) = kg m^2 s^{-2}.

3. The SI unit of power is:

(a) joule (b) newton
(c) watt (d) second

Answer: (c) watt — The watt is the SI unit of power: 1 W = 1 J s^{-1}.

4. The power of a water pump is 2 kW. The amount of water it can raise in one minute to a height of 5 metres is:

(a) 1000 litres (b) 1200 litres
(c) 2000 litres (d) 2400 litres

Answer: (d) 2400 litres — E = Pt = 2000 × 60 = 120,000 J. m = E/(gh) = 120,000/(10 × 5) = 2400 kg, approximately 2400 litres of water.

5. A bullet of mass 0.05 kg has a speed of 300 m s^{-1}. Its kinetic energy will be:

(a) 2250 J (b) 4500 J
(c) 1500 J (d) 1125 J

Answer: (a) 2250 J — K = ½mv^2 = ½ × 0.05 × 300^2 = 2250 J.

6. If a car doubles its speed, its kinetic energy will be:

(a) the same (b) doubled
(c) increased to three times (d) increased to four times

Answer: (d) increased to four times — Kinetic energy is proportional to v^2, so doubling speed gives K′ = ½$m(2v)^2$ = 4K.

7. The energy possessed by a body by virtue of its position is:

(a) kinetic energy (b) potential energy
(c) chemical energy (d) solar energy

Answer: (b) potential energy — Gravitational potential energy depends on position in a gravitational field and is mgh relative to a chosen level.

8. The magnitude of momentum of an object is doubled, the kinetic energy of the object will:

(a) double (b) increase to four times
(c) reduce to one-half (d) remain the same

Answer: (b) increase to four times — For fixed mass, K = p^2/(2m). Doubling p therefore makes kinetic energy four times as large.

9. Which of the following is not a renewable energy source?

(a) Hydroelectric energy (b) Fossil fuels
(c) Wind energy (d) Solar energy

Answer: (b) Fossil fuels — Coal, oil and natural gas are fossil fuels formed over immense geological time and consumed much faster than they form.

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B. Short Answer Questions

5.1. What is the work done on an object that remains at rest when a force is applied on it?

Zero mechanical work is done on the object because its displacement is zero: $W = F \times 0 \times \cos\theta = 0$, even if the person applying the force becomes tired and uses chemical energy internally.

5.2. A slow-moving car may have more kinetic energy than a fast-moving motorcycle. How is this possible?

Kinetic energy depends on both mass and speed: $K = \frac{1}{2}mv^2$. A car has a much larger mass, so its greater mass can more than compensate for its smaller speed and give it greater kinetic energy.

5.3. A force F_1 does 5 J of work in 10 s. Another force F_2 does 3 J of work in 5 s. Which force delivers greater power?

$P_1 = 5/10 = 0.50$ W and $P_2 = 3/5 = 0.60$ W. Therefore **F_2 delivers greater power**, although it does less total work, because it transfers energy in a shorter time.

5.4. A woman runs up a flight of stairs. The gain in her gravitational potential energy is 4500 J. If she runs up the same stairs with twice the speed, what will be her gain in potential energy?

It remains **4500 J** because the mass and vertical height are unchanged: $\Delta U = mgh$. Running faster increases power and reduces time, not the gain in gravitational potential energy.

5.5. Define work and its SI unit.

Work is the energy transferred when a force acts through a displacement: $W = Fd\cos\theta$, where θ is the angle between force and displacement. The SI unit is the **joule (J)**; 1 J is the work done by 1 N acting through 1 m in its own direction.

5.6. What is the potential energy of a body of mass m at a height h?

Near Earth’s surface, gravitational potential energy relative to the chosen ground level is **$U = mgh$**, where g is gravitational acceleration. Only changes in potential energy are physically significant unless a reference level is stated.

5.7. Find an expression for the kinetic energy of a moving body.

Consider a constant net force F accelerating mass m from rest through distance d. Then $F = ma$ and $v^2 = 2ad$.

$$\text{Work done } W = Fd = (ma)d = m(ad)$$

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Since $ad = v^2/2$, $W = \frac{1}{2}mv^2$

Therefore kinetic energy $K = \frac{1}{2}mv^2$

The work done on the body from rest becomes its kinetic energy. More generally, net work equals change in kinetic energy: $W_{net} = \Delta K$.

5.8. Define efficiency of a working system. Why can a system not have 100% efficiency?

Efficiency is useful output energy or power divided by total input, usually expressed as a percentage: η = useful output/input × 100%. A real system cannot reach 100% because some input is inevitably dissipated through friction, air resistance, electrical resistance, sound or unwanted heating; useful output is therefore less than input.

5.9. What is power? Define the unit used for it.

Power is the rate of doing work or transferring energy: $P = W/t = E/t$. Its SI unit is the **watt (W)**, where 1 watt equals 1 joule per second. Larger practical units include kilowatt (1000 W) and megawatt (10^6 W).

5.10. Differentiate between renewable and non-renewable energy sources.

Renewable sources	Non-renewable sources
Replenished naturally on a human timescale.	Finite reserves formed far more slowly than consumed.
Examples: sunlight, wind, moving water, sustainably managed biomass.	Examples: coal, petroleum, natural gas and other finite fuels.
Usually lower operating emissions, though construction and land use still have impacts.	Combustion generally releases carbon dioxide and pollutants; mining and extraction also affect the environment.

C. Constructed Response Questions

5.1. Can the kinetic energy of a body ever be negative?

No. Mass is positive and speed is squared in $K = \frac{1}{2}mv^2$, so kinetic energy is zero when the body is at rest in the chosen frame and positive whenever it moves. Velocity components may be negative, but their squares are not.

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5.2. Which one has greater kinetic energy: an object of mass m travelling with velocity v, or an object twice as heavy travelling with velocity v/2?

First object: $K_1 = \frac{1}{2}mv^2$

Second object: $K_2 = \frac{1}{2}(2m)(v/2)^2 = m(v^2/4) = \frac{1}{4}mv^2$

$K_1/K_2 = (\frac{1}{2}mv^2)/(\frac{1}{4}mv^2) = 2$

The **first object has twice the kinetic energy**. Doubling mass doubles K, but halving speed divides K by four, giving a net factor of one-half.

5.3. A car is moving along a curved road at constant speed. Does its kinetic energy change?

No. $K = \frac{1}{2}mv^2$ depends on speed, not direction, so constant mass and speed mean constant kinetic energy. Its velocity and momentum do change because direction changes, requiring an inward centripetal force.

5.4. Comment on the statement, “An object has one joule of potential energy.”

The statement is incomplete unless the type of potential energy and reference configuration are specified. Potential energy is relative: for gravitational energy, one must state the zero-height level; for elastic energy, the unstretched position is normally zero. It is meaningful to say the object’s potential energy is 1 J **relative to** that reference.

5.5. While driving on a motorway, the tyre of a vehicle sometimes bursts. What may be its cause?

High speed, friction and repeated flexing generate heat. The heated air and rubber increase tyre stress and air pressure, while overloading, under-inflation, excessive inflation, worn rubber or a cut weakens the casing. If local stress exceeds the tyre’s strength, it bursts. Correct pressure, sound tyres and safe loading reduce the risk.

5.6. While playing cricket on a street, the ball smashes a window pane. Describe the energy changes in this event.

The player’s chemical energy becomes kinetic energy of the bat and then the ball. On impact, part of the ball’s kinetic energy breaks intermolecular bonds and does work cracking and displacing the glass; the rest becomes sound, heat, vibrations and kinetic energy of fragments and the rebounding ball. Total energy is conserved, though not all remains mechanical.

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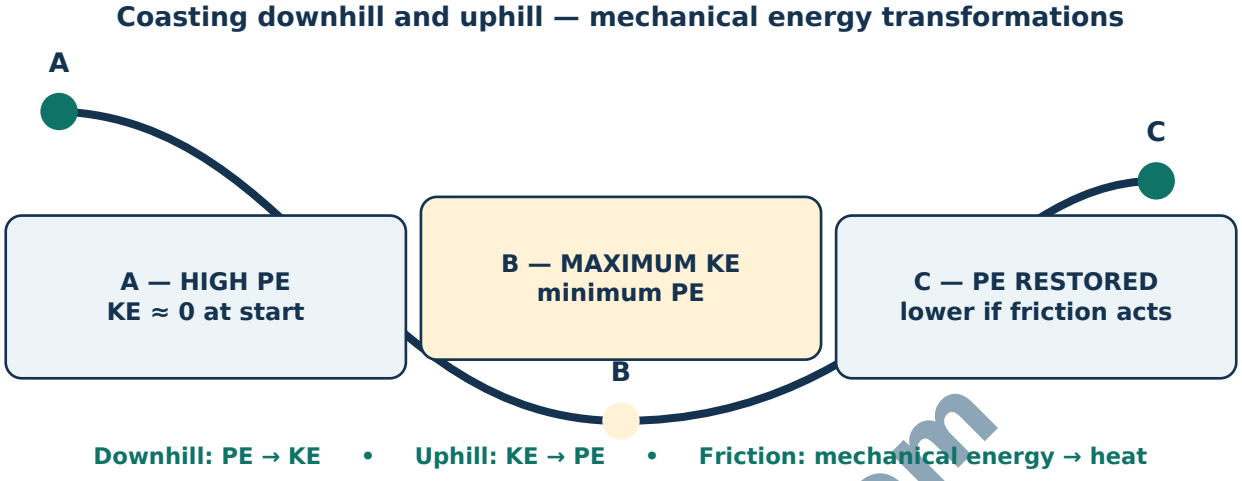
5.7. A man rowing a boat upstream is at rest with respect to the shore. Is he doing work?

The boat has zero displacement relative to the shore, so the net mechanical work done on the boat is zero. However, the rower does work on the oars and water, pushing water backward and giving it kinetic energy. His chemical energy is also converted into heat; zero net work on the boat does not mean zero energy expenditure.

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5.8. A cyclist goes downhill from the top of a steep hill without pedalling and takes it to the top of the next hill. Draw a diagram and analyse the event in terms of potential and kinetic energy.

Figure 1 — Energy labels at the two hilltops and valley



Without losses, C returns to A's height; with losses and no pedalling, C is lower.

At A, gravitational potential energy is high and kinetic energy is small or zero. Going downhill, potential energy decreases and kinetic energy increases. At B, height and potential energy are minimum while speed and kinetic energy are maximum. Climbing toward C, kinetic energy converts back to potential energy and speed falls.

Without friction or air resistance, the cyclist could regain the original height. With losses and no pedalling, some mechanical energy becomes heat and the next reachable hilltop is lower.

5.9. Is timber or wood a renewable source of heat energy? Comment.

Wood is **conditionally renewable**: trees can be replanted and regrown within a human timescale. It remains renewable only if harvesting does not exceed regrowth and forest soil and ecosystems are sustained. Deforestation makes it effectively non-renewable. Burning wood also emits smoke and carbon dioxide, so "renewable" does not mean pollution-free.

D. Comprehensive Questions

5.1. What is meant by kinetic energy? State its unit. Describe how it is determined.

Kinetic energy is the energy a body possesses because of its motion. Its SI unit is the joule (J), equal to kg m^2 s^{-2}.

For constant mass m moving at speed v, apply a constant resultant force F from rest through distance d. Newton's second law gives F = ma, and the equation of motion gives $v^2 = 2ad$.

K = work done from rest = Fd

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$$K = mad = m(v^2/2)$$

$$K = \tfrac{1}{2}mv^2$$

To determine K, measure mass in kilograms and speed in metres per second, then substitute. Net work over any interval equals the **change** in kinetic energy, W_net = K_f − K_i.

5.2. State the law of conservation of energy. Explain it with the help of a body falling from a certain height in terms of its potential and kinetic energy.

Law: Energy cannot be created or destroyed; it can only be transferred or transformed from one form to another. The total energy of an isolated system remains constant.

Drop a body of mass m from rest at height H, neglecting air resistance. At the top, U = mgH and K = 0. After falling through distance h, $v^2 = 2gh$.

$$K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}m(2gh) = mgh$$

Remaining height = H − h, so U = mg(H − h)

$$K + U = mgh + mgH - mgh = mgH = \text{constant}$$

Just before reaching the ground, U approaches zero and K approaches mgH. With air resistance, some mechanical energy becomes thermal energy and sound, but total energy still remains conserved.

5.3. Differentiate between renewable and non-renewable sources of energy. Give three examples for each.

Feature	Renewable	Non-renewable
Availability	Naturally replenished on a human timescale.	Finite stock; consumed much faster than formed.
Three examples	Solar, wind and hydroelectric energy.	Coal, petroleum and natural gas.
Supply pattern	Solar and wind vary; reservoirs and storage improve reliability.	Can provide controllable output while fuel reserves last.
Environmental effect	Usually low operating carbon emissions, but habitat, land and material impacts remain.	Extraction and combustion commonly cause pollution and substantial carbon emissions.
Long-term use	Sustainable if resource use stays within natural renewal.	Reserves deplete and eventually become uneconomic or exhausted.

Biomass is renewable only with sustainable regrowth. No energy source is entirely impact-free, so efficiency and conservation remain important.

5.4. Explain what is meant by efficiency of a machine. How is it calculated? Why is there a limit for the efficiency of a machine?

Efficiency tells what fraction of supplied energy or power emerges as useful output.

$$\eta = \text{useful energy output} \div \text{total energy input}$$

$$\text{Percentage efficiency} = (\text{useful output/input}) \times 100\%$$

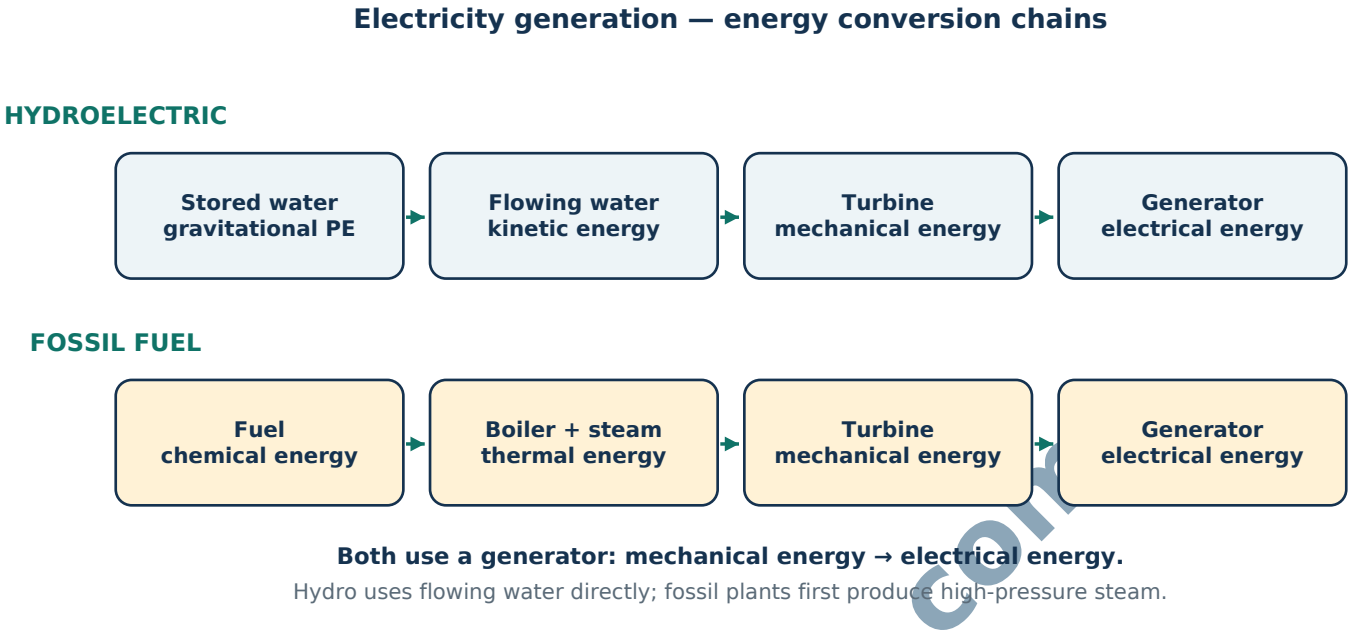
$$\text{Equivalently, } \eta = \text{useful power output} \div \text{total power input}$$

In a real machine, friction, drag, electrical resistance, vibration, deformation and sound transfer part of the input to unwanted thermal energy. Useful output is therefore less than input and $\eta < 100\%$. Lubrication, streamlined design, good insulation and efficient motors reduce losses but cannot remove every real-world dissipation.

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5.5. Describe electricity generation by drawing a block diagram for (i) hydroelectric power generation and (ii) fossil fuels.

Figure 2 — Original block diagrams for both generation routes



Hydroelectric: A dam stores water at height. Flow through a penstock converts gravitational potential energy into kinetic energy; moving water turns a turbine and the generator produces electricity.

Fossil fuel: Combustion converts chemical energy into heat. A boiler produces high-pressure steam, which turns a turbine connected to a generator. Condensers and cooling systems return water for reuse; exhaust and ash require environmental control.

E. Numerical Problems

5.1 A force of 20 N acting at an angle of 60° to the horizontal pulls a box through 3 m across a floor. How much work is done?

Given: F = 20 N; d = 3 m; θ = 60°.

$$W = Fd \cos\theta = 20 \times 3 \times \cos 60°$$

$$W = 60 \times 0.5 = 30 \text{ J}$$

Answer: 30 J.

5.2 A body moves 5 m in a straight line under an 8 N force. If the work done is 20 J, find the angle between force and direction of motion.

Given: d = 5 m; F = 8 N; W = 20 J.

$$\cos\theta = W/(Fd) = 20/(8 \times 5) = 20/40 = 0.5$$

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$$\theta = \cos^{-1}(0.5) = 60°$$

Answer: 60°.

5.3 An engine raises 100 kg of water through a height of 80 m in 25 s. What is the power of the engine?

Given: m = 100 kg; h = 80 m; t = 25 s; g = 10 m s^{-2}.

$$\text{Useful work} = mgh = 100 \times 10 \times 80 = 80{,}000 \text{ J}$$

$$P = W/t = 80{,}000/25 = 3200 \text{ W}$$

Answer: 3200 W = 3.2 kW.

5.4 A 20 kg body is at rest. A 40 N force acts on it for 5 seconds. What is its kinetic energy at the end of this time?

Given: m = 20 kg; F = 40 N; u = 0; t = 5 s.

$$a = F/m = 40/20 = 2 \text{ m s}^{-2}$$

$$v = u + at = 0 + 2 \times 5 = 10 \text{ m s}^{-1}$$

$$K = \tfrac{1}{2}mv^2 = \tfrac{1}{2} \times 20 \times 10^2 = 1000 \text{ J}$$

Answer: 1000 J.

5.5 A ball of mass 160 g is thrown vertically upward and reaches a height of 20 m. Find the potential energy gained.

Given: m = 160 g = 0.160 kg; h = 20 m; g = 10 m s^{-2}.

$$\Delta U = mgh = 0.160 \times 10 \times 20 = 32 \text{ J}$$

Answer: 32 J.

5.6 A 0.14 kg ball is thrown vertically upward with an initial velocity of 35 m s^{-1}. Find the maximum height reached by the ball.

Given: m = 0.14 kg; u = 35 m s^{-1}; at maximum height K = 0. Neglect air resistance.

$$\text{Initial } K = \tfrac{1}{2} \times 0.14 \times 35^2 = 85.75 \text{ J}$$

At maximum height: $mgh = 85.75$

$$h = 85.75/(0.14 \times 10) = 61.25 \text{ m}$$

Equivalent: $h = u^2/(2g) = 35^2/20 = 61.25$ m

Answer: 61.25 m above the launch point.

5.7 A girl on a swing is 1.2 m above ground at the lowest point and 2.0 m at the highest point. What is her maximum velocity and where?

Given: $\Delta h = 2.0 - 1.2 = 0.8$ m. Maximum speed occurs at the lowest point.

Loss of PE = gain of KE: $mg\Delta h = \frac{1}{2}mv^2$

$v = \sqrt{(2g\Delta h)} = \sqrt{(2 \times 10 \times 0.8)} = \sqrt{16} = 4$ m s^{-1}

Answer: 4 m s^{-1} at the lowest position.

5.8 A person pushes a lawn mower with a force of 50 N at 45° to the horizontal. If it moves 20 m, how much work is done?

Given: F = 50 N; d = 20 m; θ = 45°.

$W = Fd \cos \theta = 50 \times 20 \times \cos 45°$

$W = 1000 \times 0.7071 = 707.1$ J

Answer: Approximately 707 J.

5.9 Calculate work done in (i) pushing a 5 kg box up a frictionless 10 m incline at 30° to horizontal and (ii) lifting it vertically to the same top.

Given: m = 5 kg; L = 10 m; θ = 30°; vertical height $h = L \sin \theta = 5$ m.

(i) Force parallel to plane at constant speed = $mg \sin 30° = 5 \times 10 \times 0.5 = 25$ N

$W_1 = 25 \times 10 = 250$ J

(ii) $W_2 = mgh = 5 \times 10 \times 5 = 250$ J

Answer: (i) 250 J (ii) 250 J. Gravitational work depends on vertical height, not path length.

5.10 A 10 kg box is pushed up a 15 m ramp with an 80 N force and rises 5 m. What is the efficiency?

Given: m = 10 kg; F = 80 N; L = 15 m; h = 5 m.

Input work = $FL = 80 \times 15 = 1200$ J

Useful output = mgh = 10 × 10 × 5 = 500 J

η = 500/1200 × 100% = 41.67%

Answer: Efficiency ≈ 41.7%; about 58.3% is dissipated or otherwise not useful lifting work.

5.11 A force of 600 N pushes a box 5 m in 15 s. Calculate the power.

Given: F = 600 N; d = 5 m; t = 15 s; force acts along motion.

W = Fd = 600 × 5 = 3000 J

P = W/t = 3000/15 = 200 W

Answer: 200 W.

5.12 A 40 kg boy runs upstairs 10 m high in 8 s. What power does he develop?

Given: m = 40 kg; h = 10 m; t = 8 s; g = 10 m s^{-2}.

Gain in PE = mgh = 40 × 10 × 10 = 4000 J

P = E/t = 4000/8 = 500 W

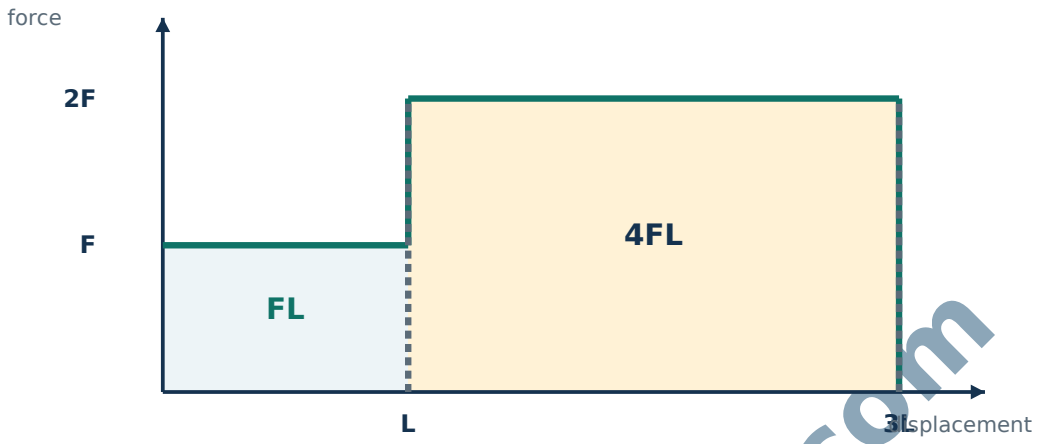
Answer: 500 W of useful climbing power.

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5.13. A force F acts through distance L on a body. The force is then increased to 2F and acts through a further distance 2L. Sketch a force–displacement graph and calculate total work.

Figure 3 — Two rectangular areas represent the two stages of work

Work as the area under a force–displacement graph



Total work = area = FL + (2F)(2L) = FL + 4FL = 5FL

$$W_1 = F \times L = FL$$

$$W_2 = 2F \times 2L = 4FL$$

$$W_total = FL + 4FL = 5FL$$

Total work = 5FL, represented by the total shaded area.

F. More Important Questions (from the text)

1. When is work positive, negative or zero? Give examples.

Work	Condition	Example
Positive	$0° \le \theta < 90°$; force has a component along displacement.	Pulling a cart forward; gravity on a falling ball.
Zero	$d = 0$ or $\theta = 90°$; no displacement or no force component along it.	Holding a bag stationary; centripetal force in uniform circular motion.
Negative	$90° < \theta \le 180°$; force component opposes displacement.	Friction on a sliding box; gravity while a ball rises.

2. How are work, energy and power related?

Energy is the capacity to do work. Doing work transfers energy, and the work–energy theorem states W_net = ΔK. Lifting at constant speed transfers energy into gravitational potential energy.

Power tells how quickly that transfer occurs: P = W/t. Two machines can perform the same work, but the one taking less time has greater power. Energy is measured in joules, whereas power is measured in watts.

Points to Remember

Work: W = Fd cos θ; 1 J = 1 N m = 1 kg m² s⁻².

Kinetic energy: K = ½mv² = p²/(2m); never negative for ordinary mass.

Gravitational PE: U = mgh relative to a selected zero level.

Conservation: energy transforms or transfers; total energy of an isolated system is constant.

Mechanical energy: K + U remains constant only when nonconservative losses are negligible.

Power: P = W/t = E/t; 1 W = 1 J s⁻¹.

Efficiency: η = useful output/input; percentage efficiency multiplies by 100%.

Scaling: doubling speed makes K four times; doubling mass makes K twice.

Graphs: area under a force–displacement graph equals work.

Sources: solar, wind and hydro replenish; coal, oil and gas are finite.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	a	2	d	3	c	4	d	5	a
6	d	7	b	8	b	9	b		

How to attempt this chapter in the paper

• Use only the force component parallel to displacement for work; vertical height, not ramp length, determines mgh.
• Convert grams to kilograms, kilowatts to watts and minutes to seconds before substitution.
• For power, calculate work or energy first and then divide by total time.
• At the highest point of vertical flight, kinetic energy is zero but gravitational acceleration is not.
• Efficiency compares useful output with total input; do not reverse the ratio.
• A renewable label does not automatically mean impact-free; mention conditions and losses where relevant.

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